Norwegian University
of Life Sciences

TEL200 – Introduction to Robotics

David A. Anisi

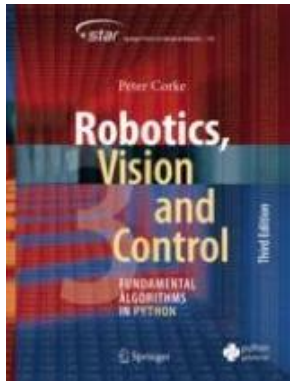
Lecture 1: Introduction and overview

TEL200 Introduction to Robotics

- **TEL200 is a project-/ gathering-based course**
- **Lectures/Exercises:** Three gatherings (Samling 1-3) TF4-102
- **Project/lab:** Tuesdays 14.15-18.00 TF4-102
  Thursdays 12.15-14.00 TF4-102
- **Teachers:**
  - david.anisi@nmbu.no (Course responsible, lectures)
  - ilwaad.hassan.mahamed@nmbu.no (YuMi project)
  - ludvik.hoibjerg-aslaksen@nmbu.no (Excercises + WalkingRobot)
  - mikkel.einvik.stryker@nmbu.no (Lab. engineer)

TEL200 Introduction to Robotics

- Course content
  - Coordinate systems
  - Time and motion
  - Robot programming (RS/Python)
  - Sensors, perception, vision & robot learning
  - Manipulator- and differential kinematics
- Course content contd.
  - Mobile robots
  - Navigation
  - Localization and mapping
- Course material
  - RVC 3rd Ed. + QUT Robot Academy + lecture notes + files on Canvas



TEL200 Introduction to Robotics

Tell me and I will forget.
Show me and I will remember.
Involve me and I will understand.
– Chinese proverb

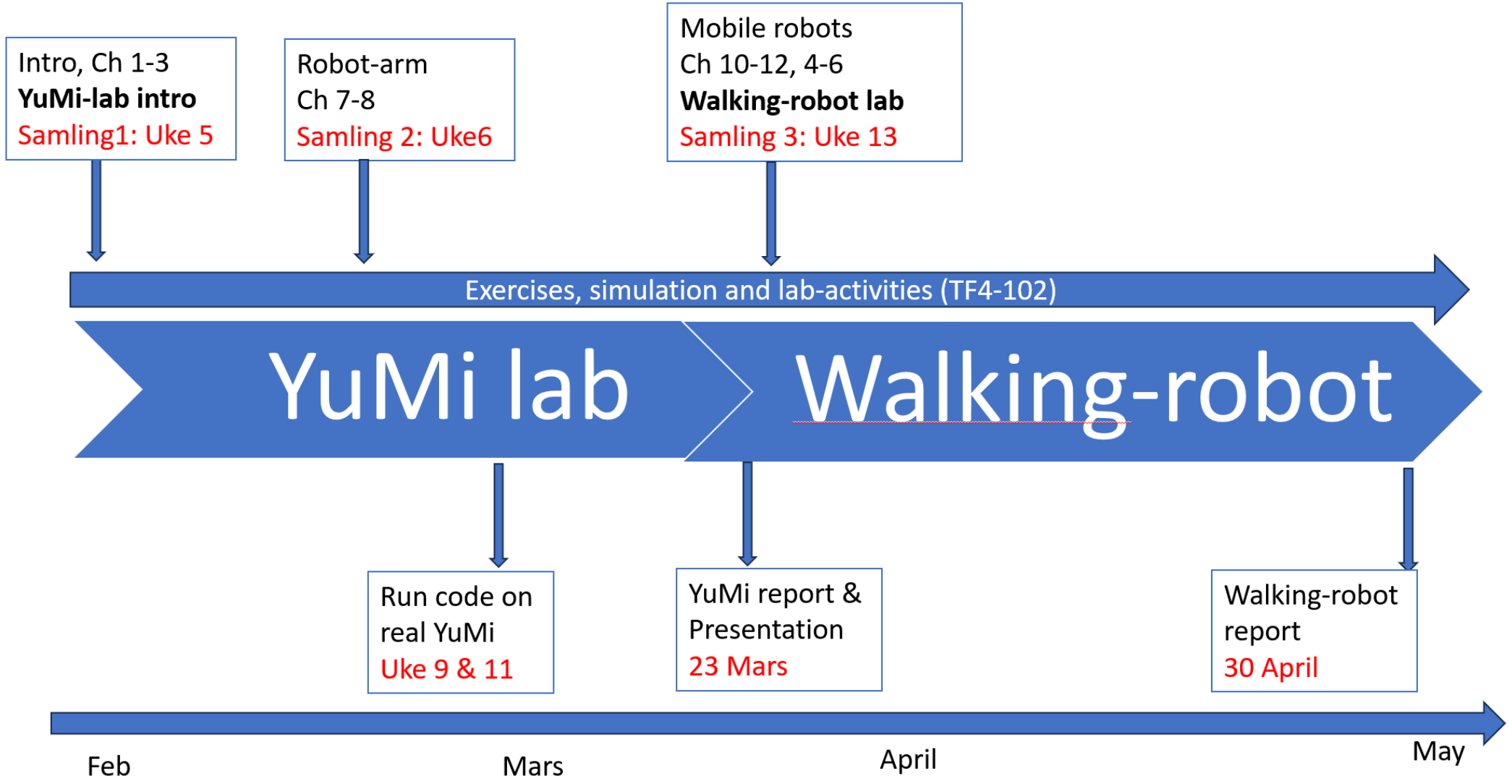
- Learning outcomes
  1. Wide introduction and peek into the magic world of Robotics!!
  2. Develop a broad understanding of fundamental principles and problems in Robotics and provide a foundation for further studies of the different topics in greater depth in other, more specialized courses (T-competancy)
  3. Acquired knowledge about, and skills in, design, analysis and the application of robots
  4. Combine theory with practice: experiment with programming of real robots
  5. Understand basic concepts of Robotic Vision and Learning

TEL200 Introduction to Robotics

- Learning activities and assessments (nominal workload, approx. 250 hrs.)
  - Lectures and exercises in accordance with detailed schedule (see also Canvas)
  - Self-study (reading, videos, pods)
  - **No Exam!** Final grade (A-E/F) based on the following **mandatory deliveries**

Module	Type	Starts	Due	Delivery requirements	Grade
RobotStudio + YuMi lab (60%)	Oral Presentation (10 min)	29 Jan.	23 Mars	PPT files & videos (max 3 min)	P/F
	File submission	29 Jan.	23 Mars	PDF report (max 10 pages) + video (3 min) + RS Pack&Go	0-60 p
Walking Robot (40%)	File submission	24 Mars	30 April	PDF report (max 10 pages) + video (3 min) + py-files	0-40 p

Course time-line



Course schedule

	Samling 1 Chapters 1-3		Samling 2 Chapters 7-8			Samling 3 Ch. 4-6, Robot learning & vision				
Date	28.jan	29.jan	03.feb	04.feb	05.feb	23.mar	24.mar	25.mar	26.mar	27.mar
Day\ Time	Wednesday	Thursday	Tuesday	Wed	Thursday	Monday	Tuesday	Wed	Thursday	Friday
08.00-9.00	Ch. 1									Ex. Ch 5-6
09.00-10.00	Ch. 1 + YuMi									Ex. Ch 5-6
10.00-11.00		RS Intro	Ch. 7		Ex. Ch. 7-8		Rob Vision& Learning		Proj. Lab	
11.00-12.00		RS Intro	Ch. 7		Ex. Ch. 7-8		Ch. 4 + proj intro		Proj. Lab	
12.00-13.00		Ex. Ch. 2-3	Ch. 7		Ex. Ch. 7-8		Ch. 4		Ch. 6	
13.00-14.00	Ch. 2	Ex. Ch. 2-3	Ch. 7		Ex. Ch. 7-8	Oral	Ch. 4		Ch. 6	
14.00-15.00	Ch. 2		YuMi lab			Presentation	Ex. Ch4/Walking Rob	Ch. 5	Ch. 6	
15.00-16.00	Ch. 3		YuMi lab			TF1-220 Delta	Ex. Ch4/Walking Rob	Ch. 5	Ch. 6	
16.00-17.00	Ch. 3		YuMi lab	Ch. 8		Sensors & Perc.	Ex. Ch4/Walking Rob	Ch. 5		
17.00-18.00	Ch. 3		YuMi lab	Ch. 8		Sensors & Perc.	Ex. Ch4/Walking Rob	Ch. 5		

Agenda

First hour

- Course overview and info
- What is a robot? (discussion groups)
- Brief history of robotics (Ch. 1)
- Evolution of selected robotic sub-domains (videos)

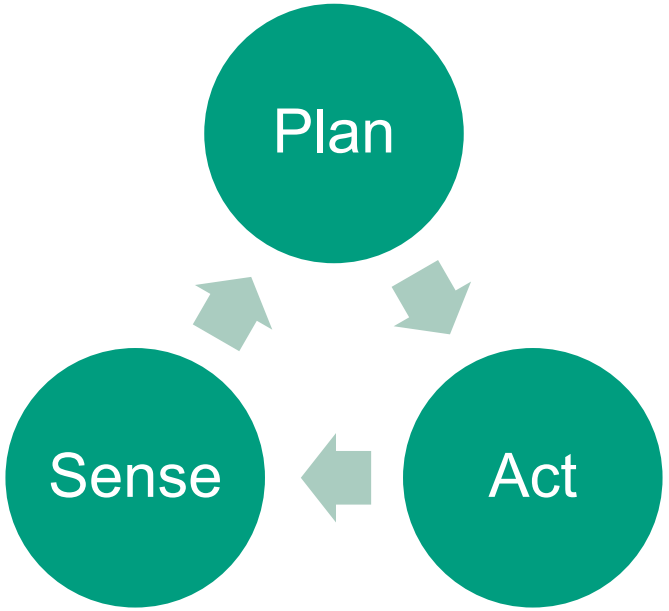
Second hour

- Grand challenges and fundamental problems in robotics -> course content
- Ethics in robotics (discussion groups)
- State of the art (videos)

What is a robot?

What is a robot?

- A goal-oriented machine that can sense, plan and act (autonomously)
- Exteroceptive- and proprioceptive sensors



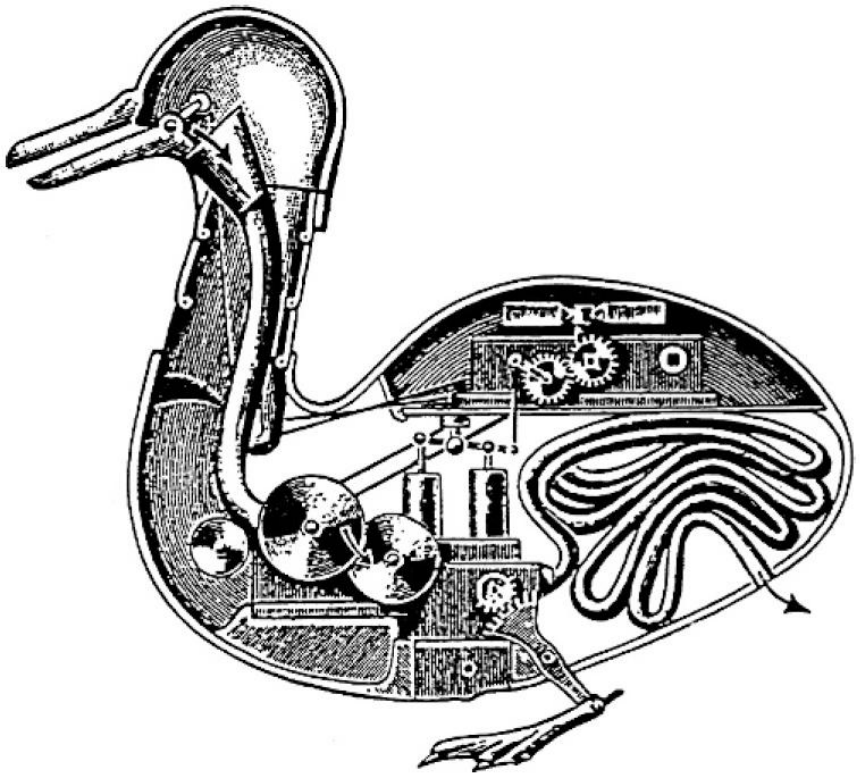
- "Robot" originates from 1920 Czech sience fiction play “Rossum’s Universal Robots” (R.U.R), by Karel Čapek.

Brief history of robotics

- Karakuri Ningyo, Japan (1700s ~)
- Traditional Japenese mechanized automata

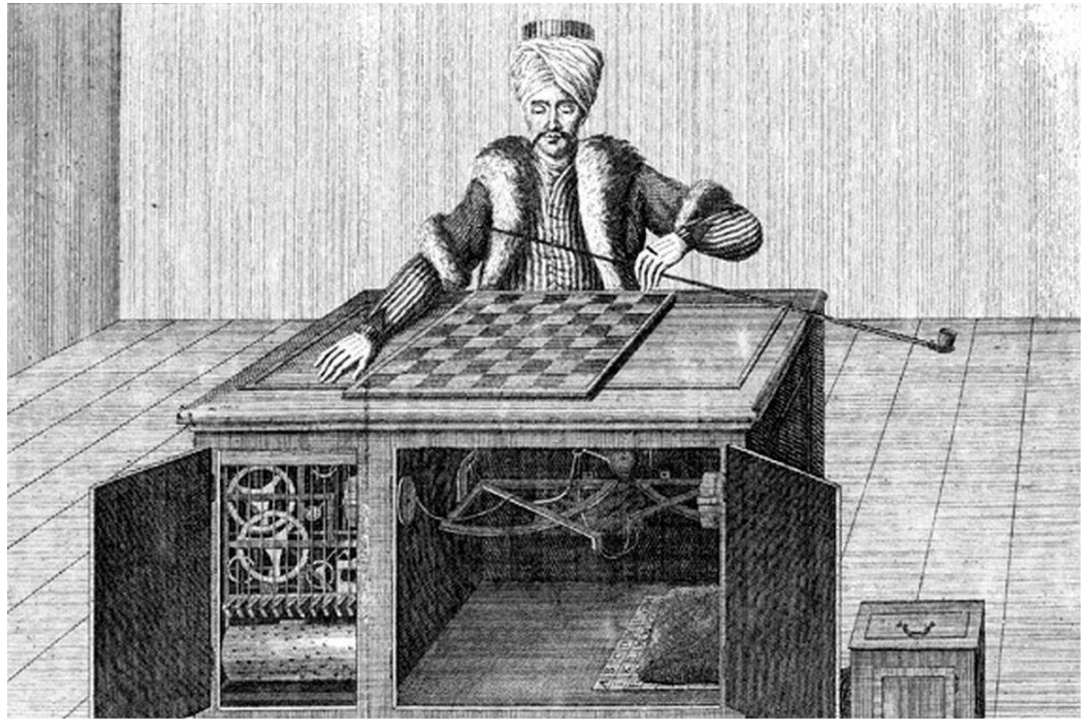
Brief history of robotics

- Vaucanson's duck (1739)
- Early programmable machine
- Automaton that could flap its wings, eat grain and defecate.



Brief history of robotics

- Mechanical Turk, Hungary (1770s ~)
- Early programmable machine
- Automaton Chess Player (fake)
- Won against Napoleon Bonaparte in 1809
- See video



Brief history of robotics

- The Jacquard loom (1801)
- Re-programmable machine
- Program stored on punched cards

Brief history of robotics

- Developed by Raymond Goertz in 1954
- Used for remote handling of hazardous material
- Teleoperation: Master-slave
- Viewed the operation through lead-glass window or via a television link
- Force feedback

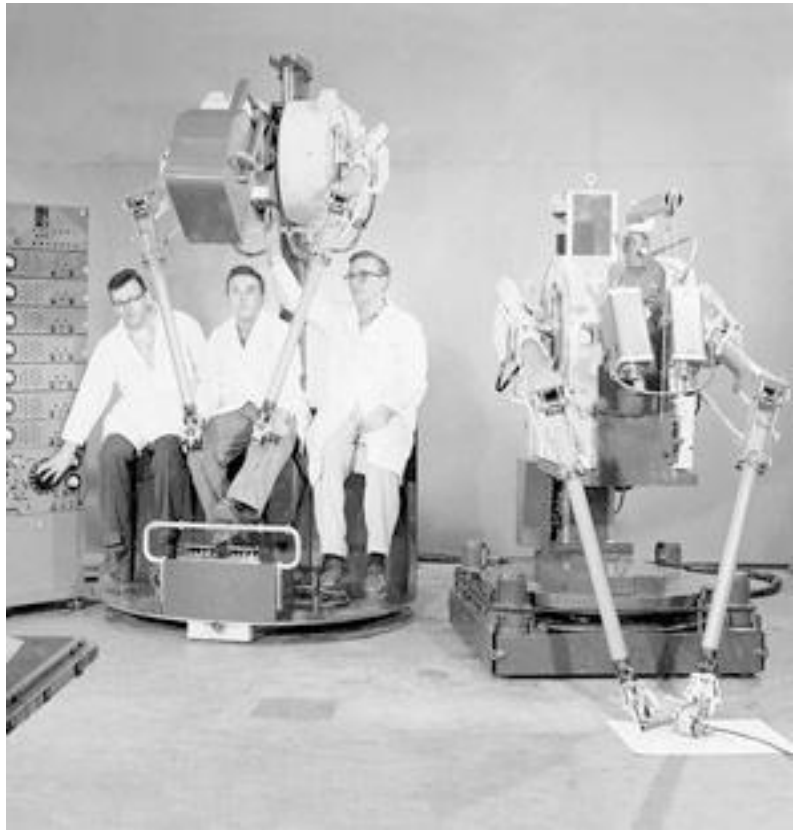


Image courtesy: Argonne National Laboratory

Brief history of robotics

- Unimation Inc.: worlds first robotics company
- Founded by George Devol and Joseph Engelberger in 1956
- Puma (**P**rogrammable **U**niversal **M**achine **M**anipulation **A**rm), based on Stanford Arm (1969) is the first robotic manipulator arm with a closed form for inverse kinematics

Brief history of robotics

- Unimation Inc.: worlds first robotics company
- Founded by George Devol and Joseph Engelberger in 1956

George Devol and Joseph F. Engelberger at a cocktail party in 1956:

GD: 50 percent of the people who work in factories are really putting and taking.

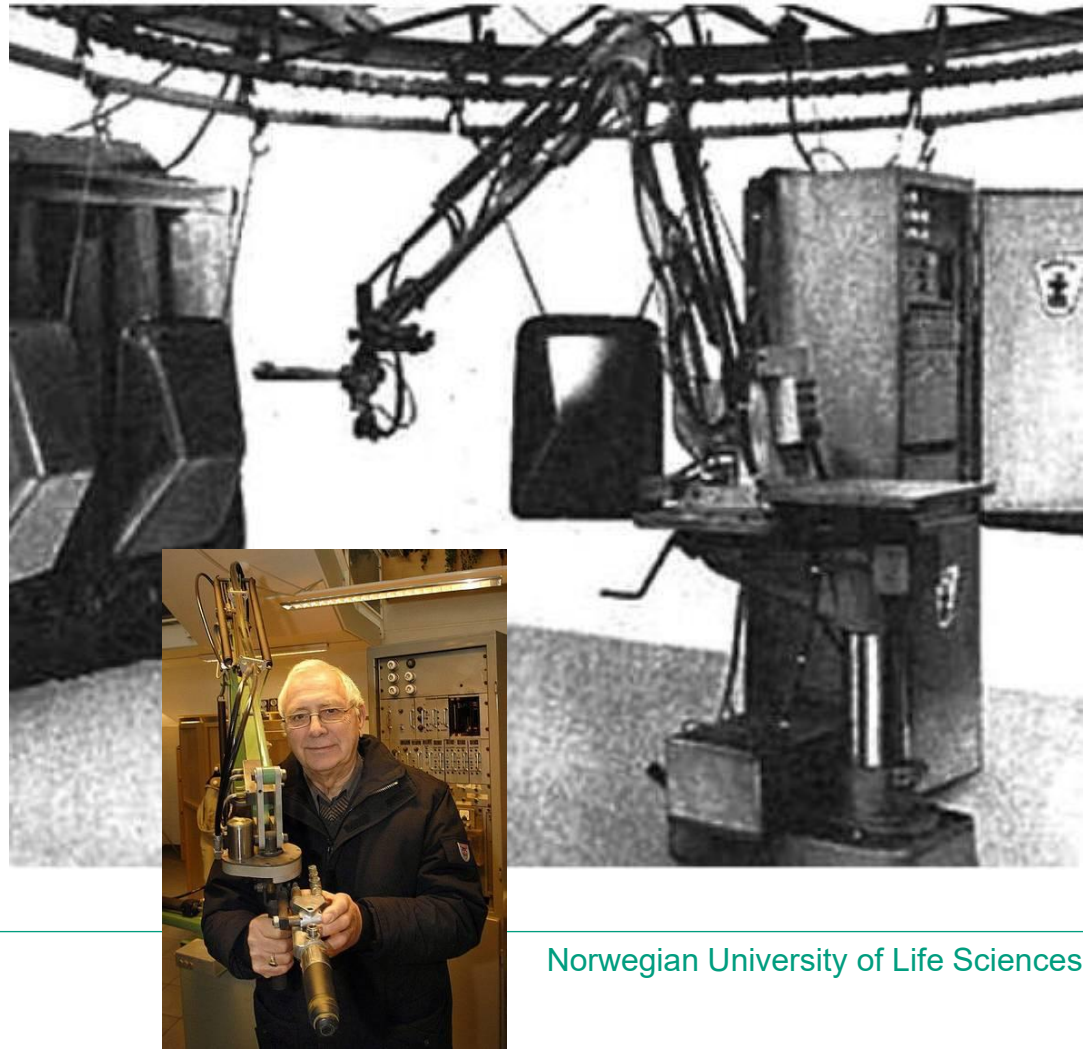
JE: Why are machines made to produce only specific items?

GD: How about approaching manufacturing the other way around, by designing machines that could put and take anything?

Unimation, Inc., the world's first robot company, was formed as a result of this discussion.

Brief history of robotics

- Worlds first painting robot
- ABB Robotics (Trallfa), 1964- 1969
- Developed by Ole Molaug and collegues from Bryne, Norway.
- Molaug also invented world's first centralized feeding system for farmed fish in 1979

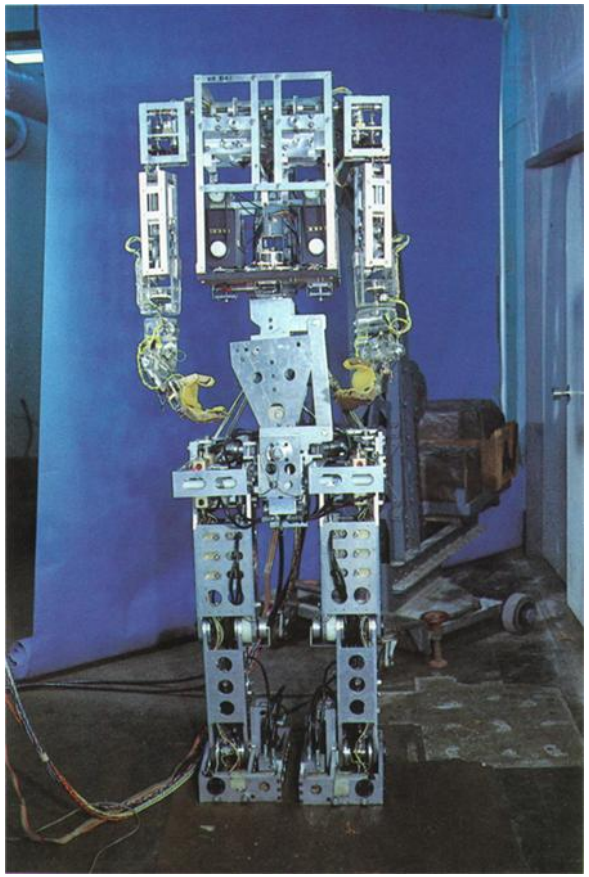


Brief history of robotics

- World's first commercially available all electric micro-processor controlled robot
- IRB6, ABB Robotics (ASEA), 1973

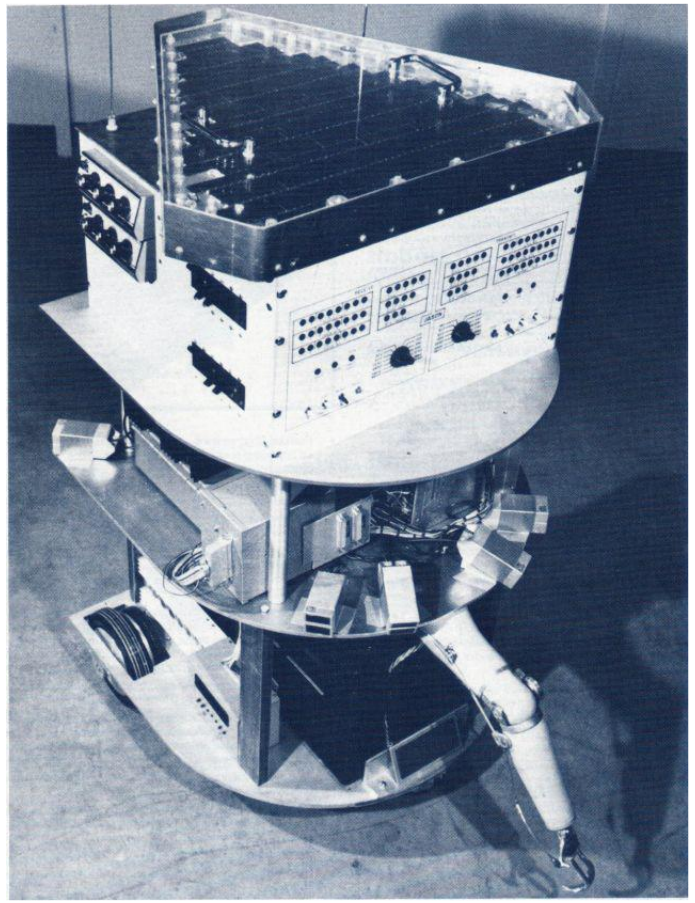
Brief history of robotics

- Wabot
- Waseda (1973)



Brief history of robotics

- Berkeley 'Jason'
- Mobile autonomous robot (1974)



Brief history of robotics

- Nintendo 'Chiritorii' (1979)
- Remote controlled vacuum clearner
- Opens up service-robot market

Brief history of robotics

- Omnidirectional robot (1980)

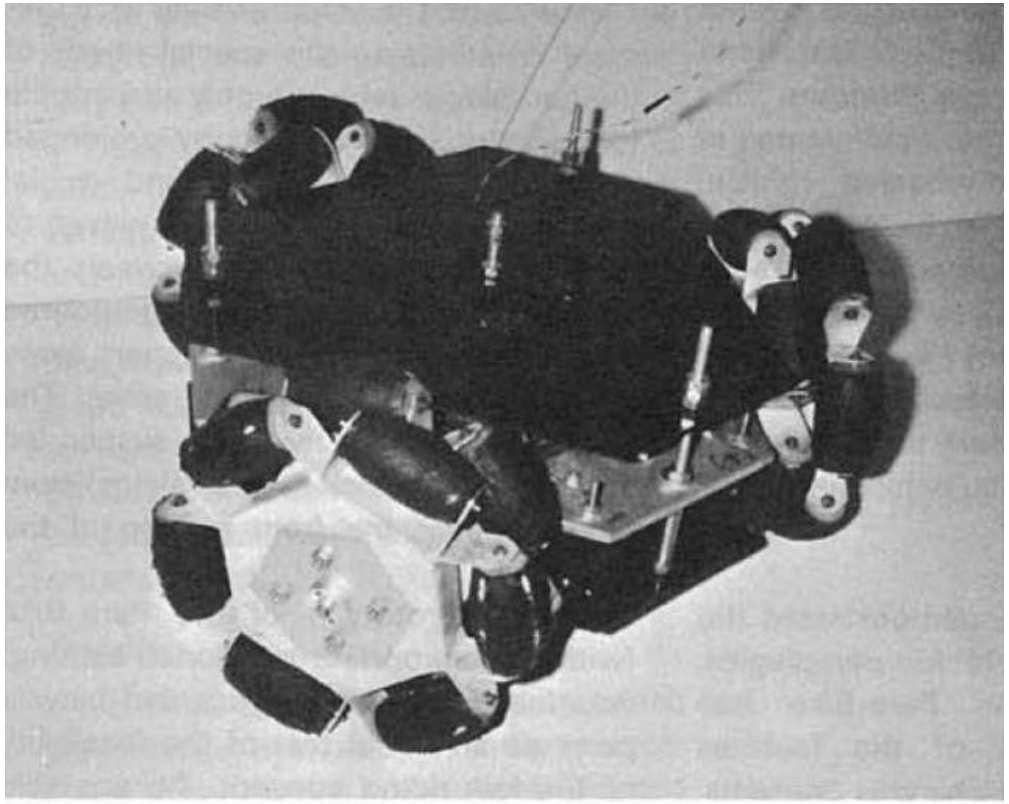


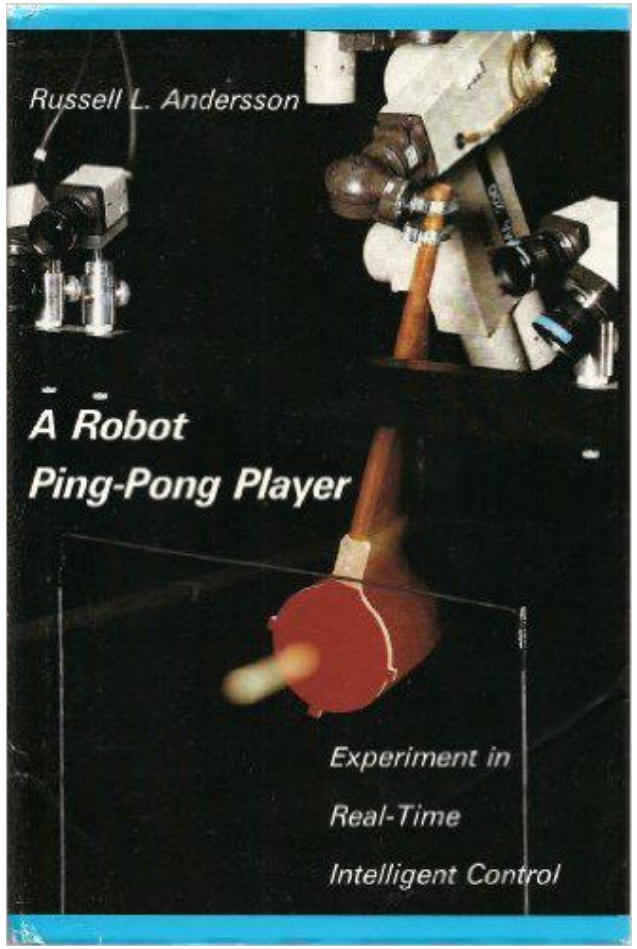
FIGURE 7.
Scale model of omnidirectional mobility base.

Brief history of robotics

- Ernst Dickmanns (VaMoRs), 1986, vision-based autonomous (95%) driving

Brief history of robotics

- Robot table tennis, Russell Andersson (MIT), 1988



Brief history of robotics

- Electrolux 'Trilobite', 1997

Brief history of robotics

- Draganflyer, Christine and Zenon Dragan (1999)

Brief history of robotics

- Cobots: safe, light-weight, F/T sensor
- DLR Light-weight robots, 1991-
- Realizing dream of 1:1 payload to mass ratio

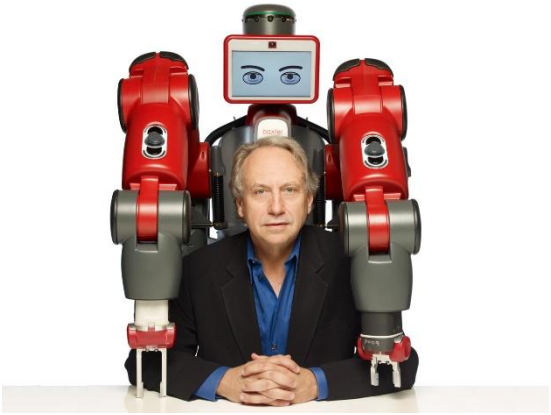
I: 1991-1996

II: 1998-2002

III: 2003 (14/14)

Kuka LBR (2004)

Universal Robotics UR5 (2008)



Rethink Robotics, Baxter (2012)

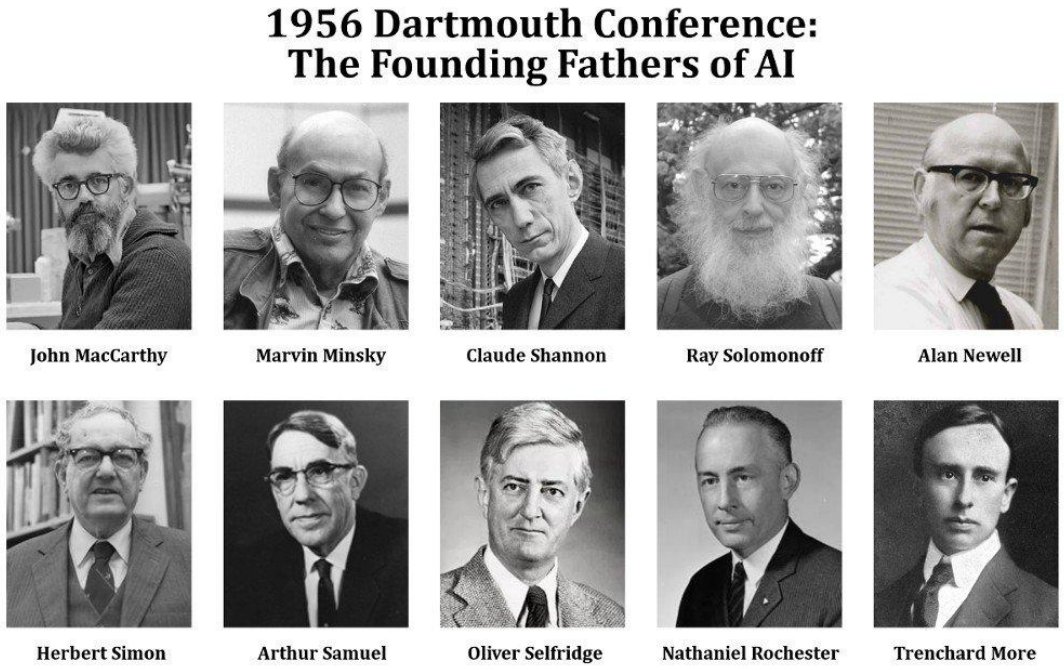
ABB Yumi (2015)

Robot taxonomy

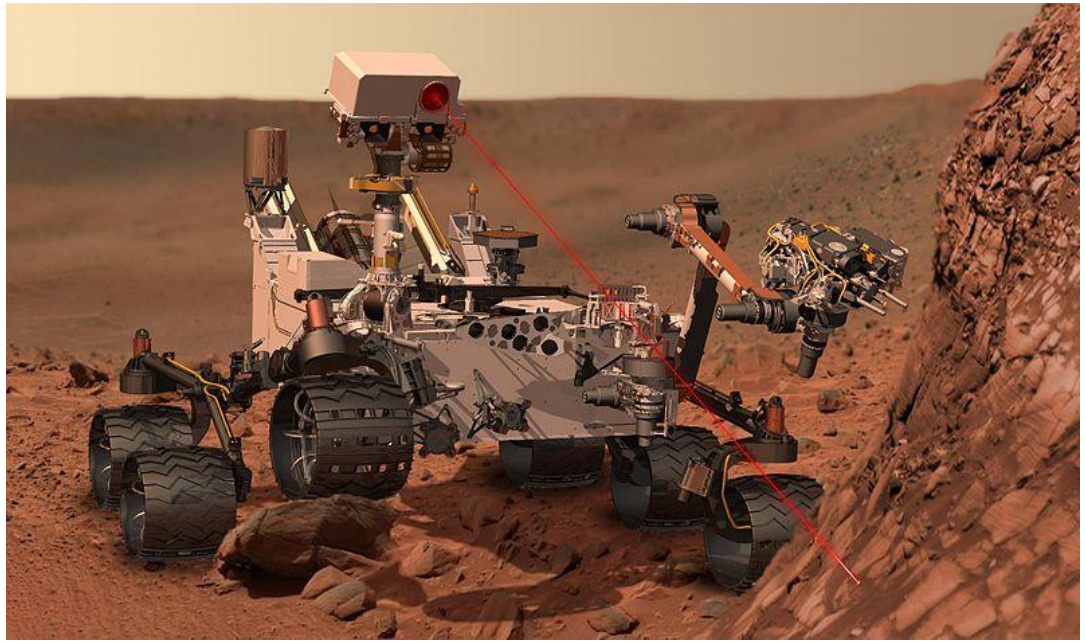
- Stationary - Mobile
- Teleoperation (ROV) – Autonomous (AV)
- Drones: Fixed wing – Quadrotors
- Field Robots

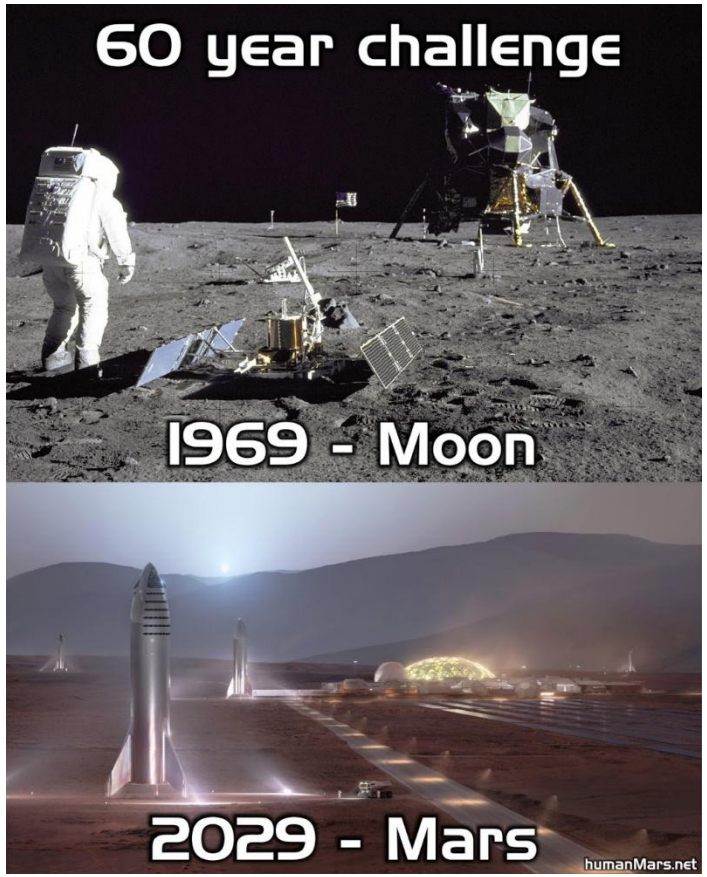
  Aerial – Ground - Surface – Underwater
- Manufacturing – Service
- Robotic Manipulator - Cobots
- Humanoids

50 year challenge I
Birth of AI (1956) to AlphaGo (2015)



50 year challenge II
Man on the moon (1969) to Curiosity Rover (2013) or Mars (2029)?





50 year challenge III ?
Robotic Agriculture

50 year challenge IV
RoboCup

By mid-21st century, a team of fully autonomous humanoid robot soccer players shall win the soccer game, complying with the official rule of the FIFA, against the winner of the most recent World Cup (1997)

50 year challenge IV?
RoboCup

Robotic Evolution – Humanoid - Asimo

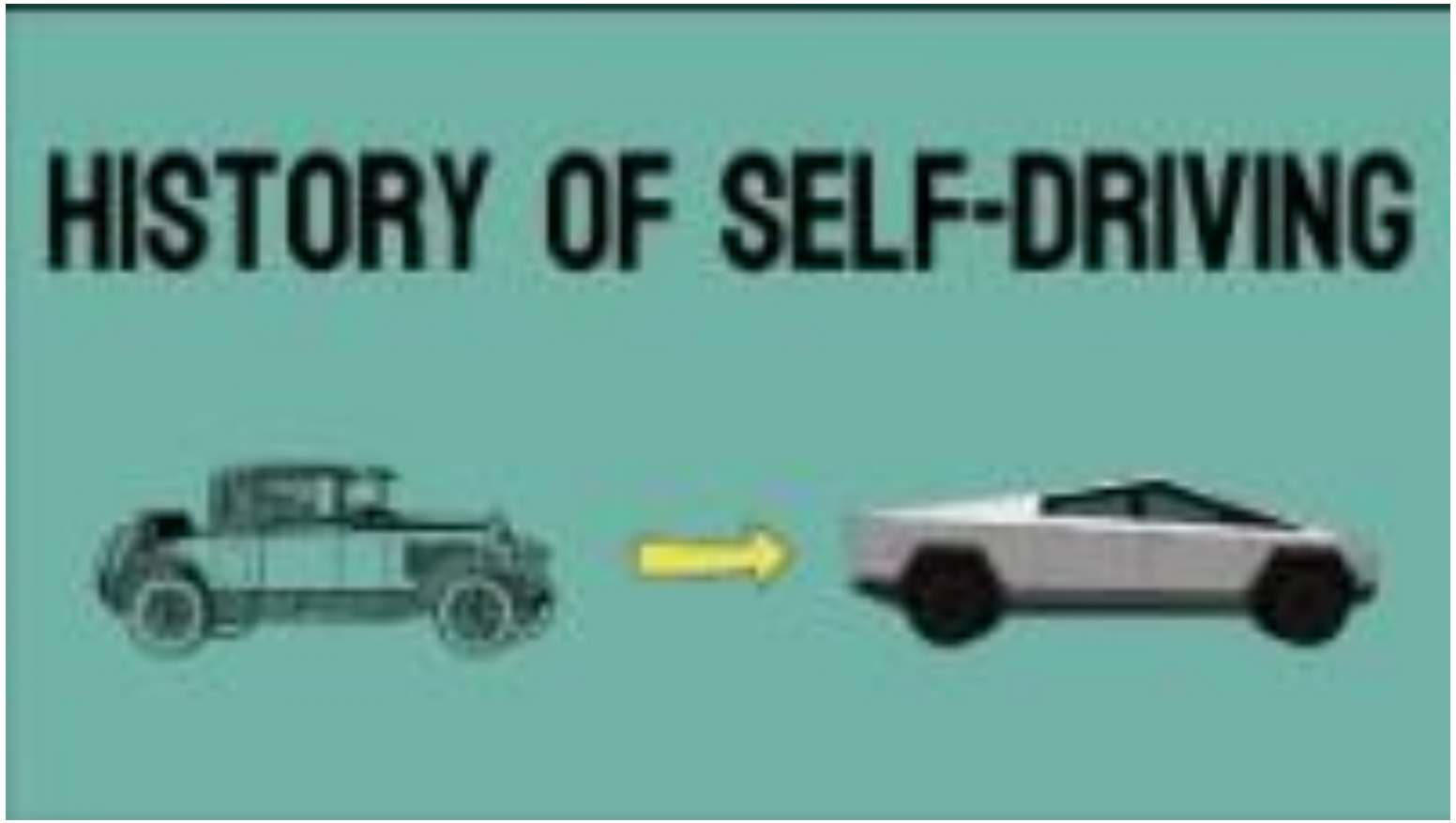
Robotic Evolution – Humanoid - Atlas

Hydraulic Atlas (until 2024)

All new full electric Atlas (2024)

Robotic Evolution – Drones

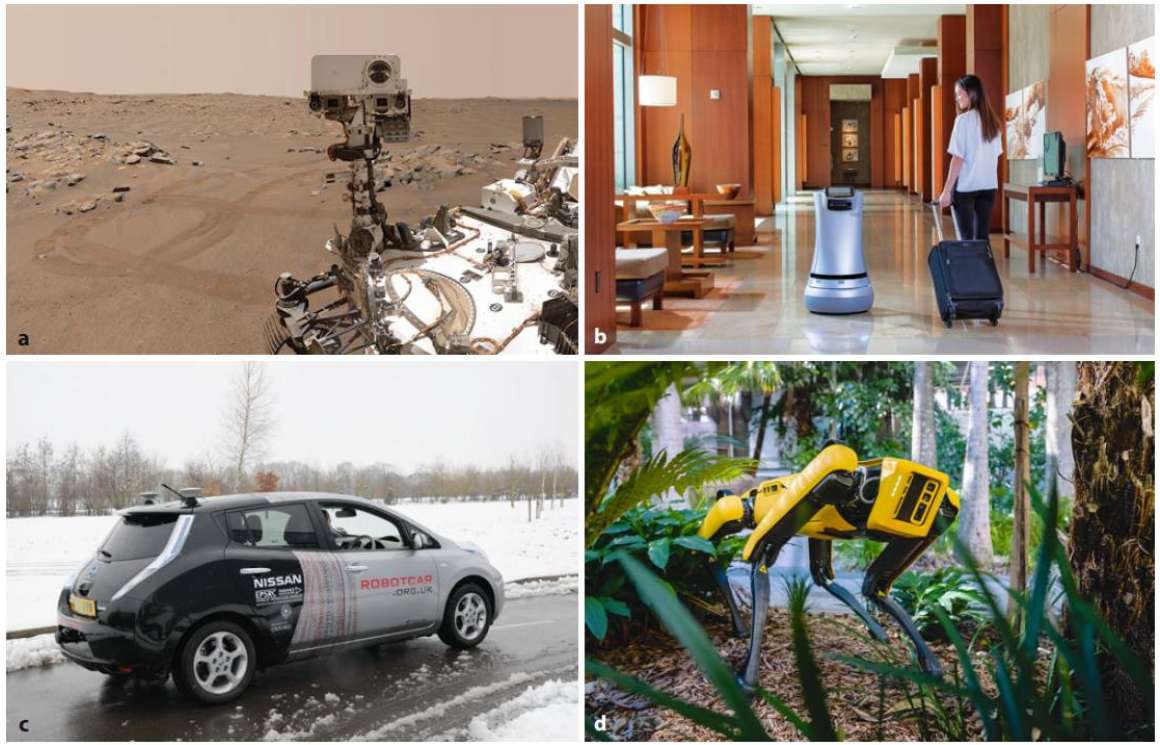
Robotic Evolution – Self-driving cars



Robotic Evolution –
DARPA Grand Challenges and Self-driving cars

Grand challenges of robotics

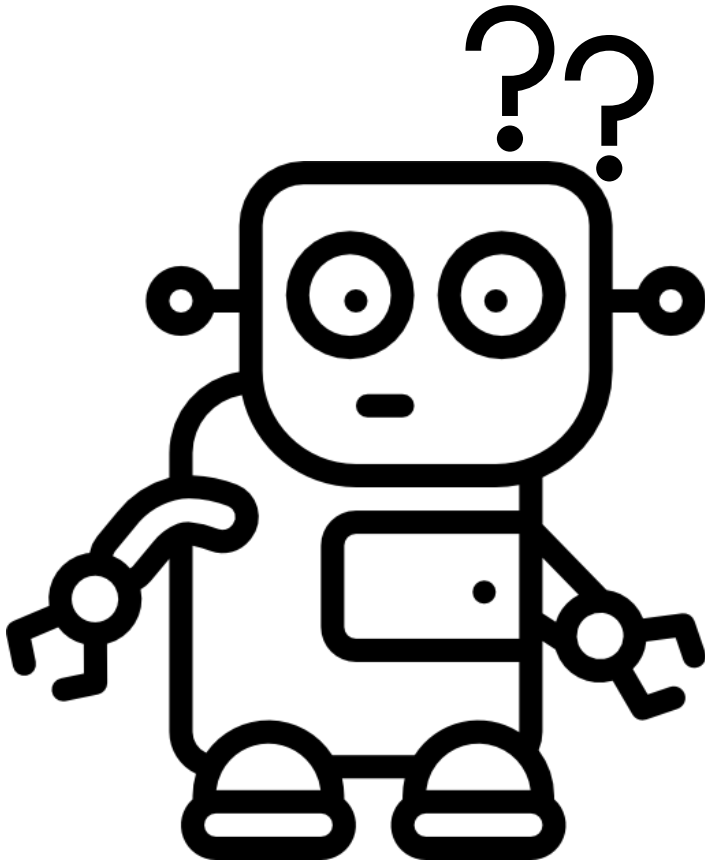
- Field- and service robots face specific and significant challenges
  - Complex, cluttered and changing environment
  - Risk of harm and safety
  - Sensors and perception
  - Actuation and manipulation



Fundamental questions in Robotics

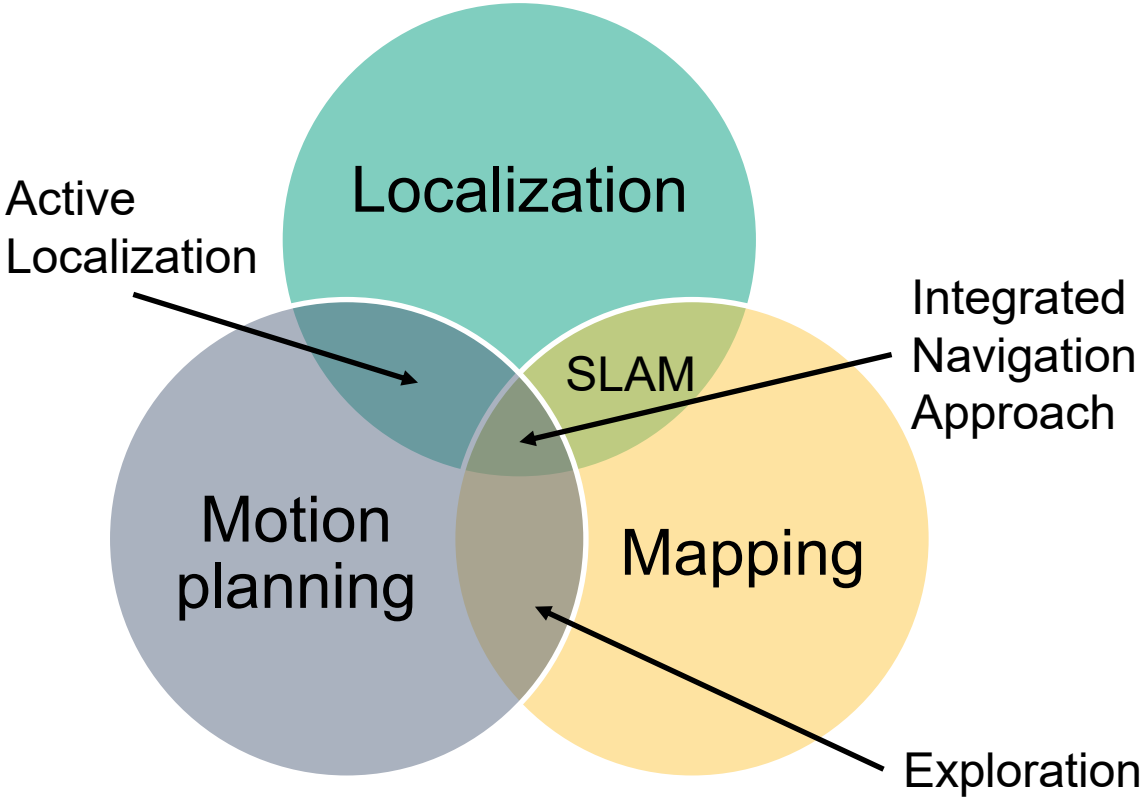
- Where should I go? What to do there?
  - Task planning
- Where am I?
  - Localization
- What is the surrounding environment?
  - Mapping
- How do I move safely from A to B?
  - Motion planning

Robot Navigation

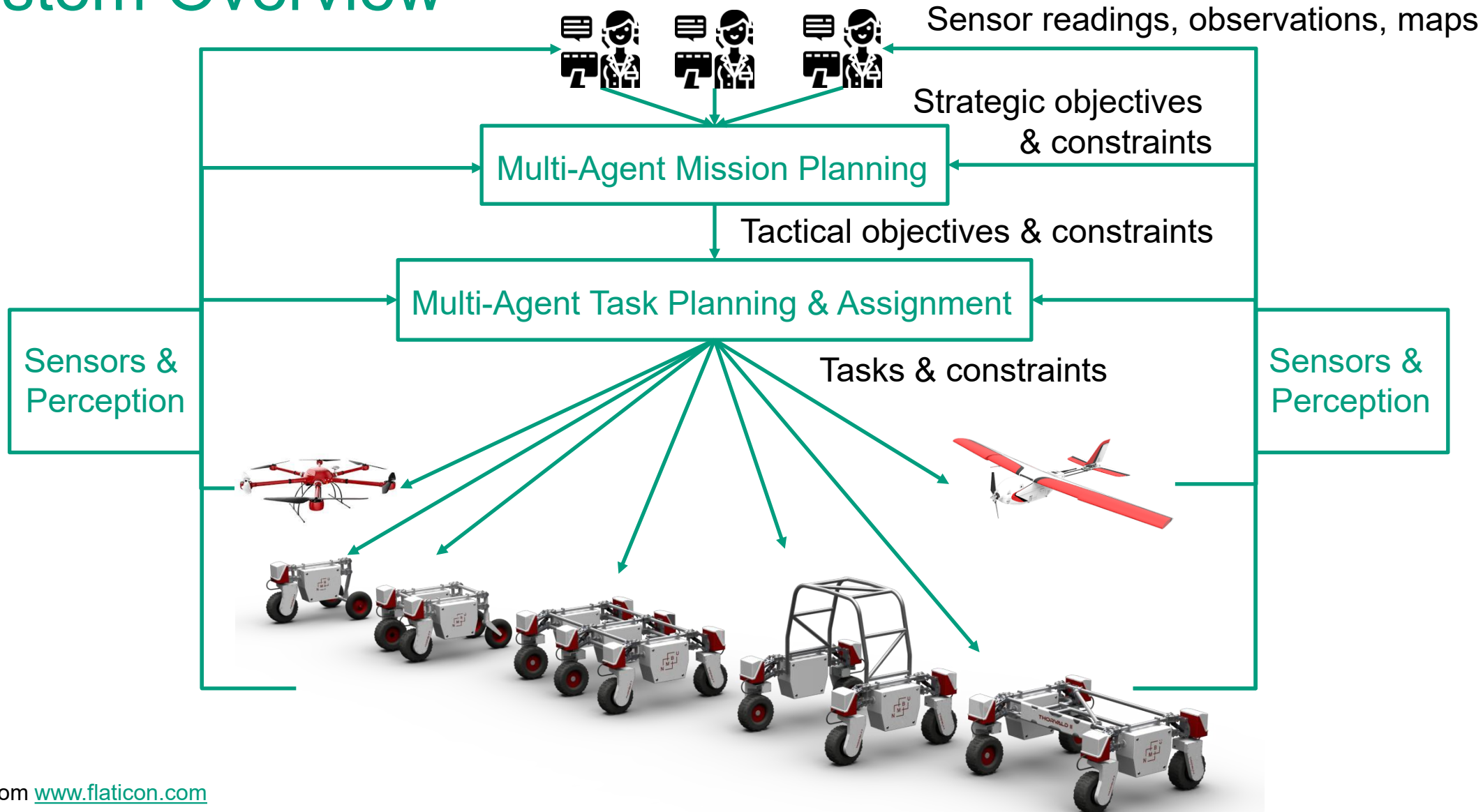


Robot Navigation

- «*the process of directing a vehicle so as to reach the intended destination*”
  - IEEE Standard 172-1983

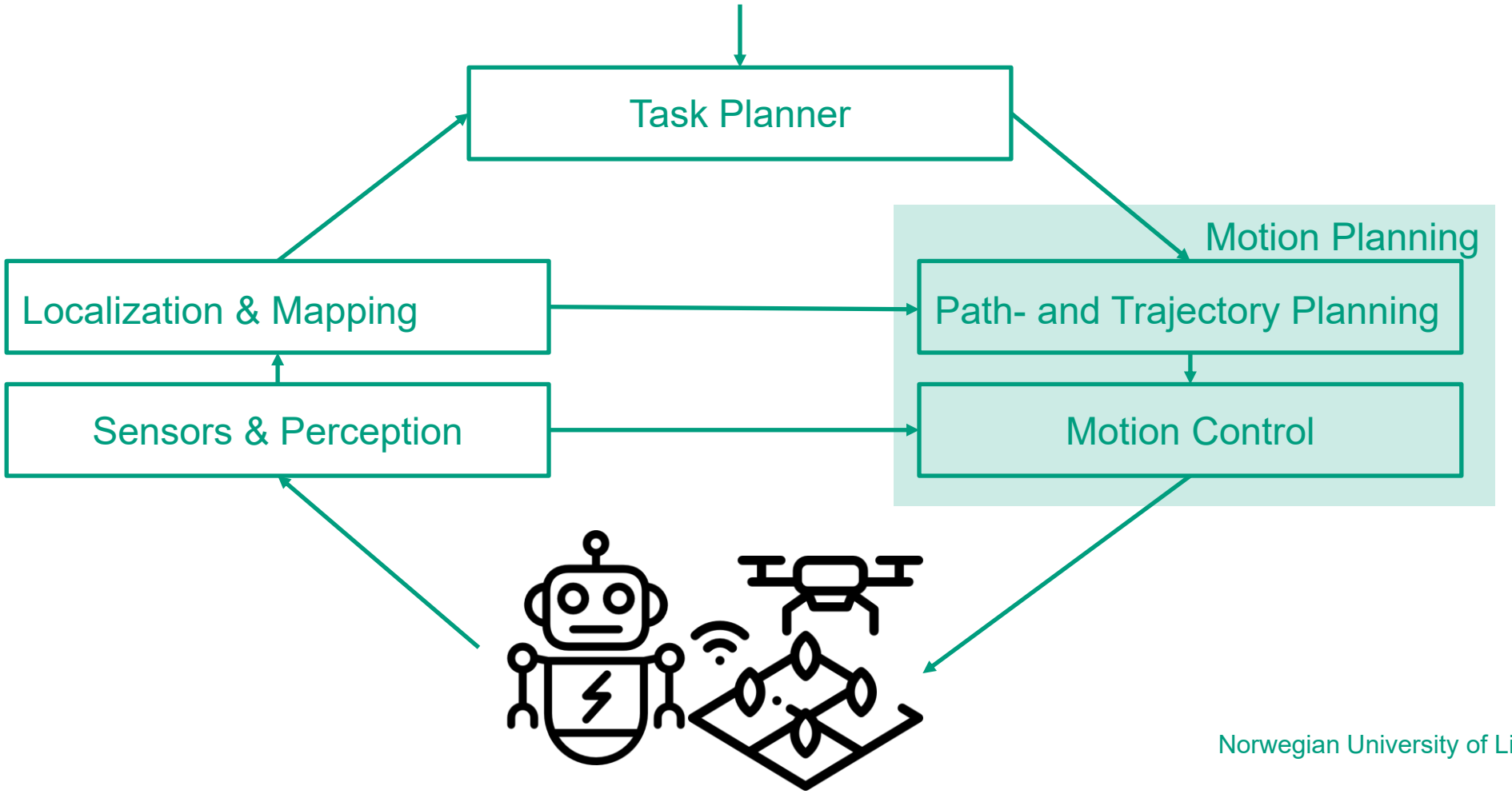


System Overview



Icons from www.flaticon.com

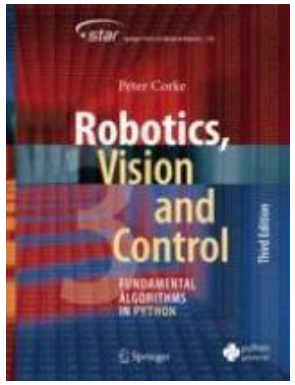
System overview



TEL200 Introduction to Robotics

- Course content
  - Coordinate systems, Ch2.
  - Time and motion, Ch3.
  - Robot programming (RS/Python)
  - Sensors, perception, vision & robot learning
  - Manipulator- and differential kinematics, Ch. 7-8

- Course content contd.
  - Mobile robots, Ch. 4
  - Navigation, Ch. 5
  - Localization and mapping, Ch. 6

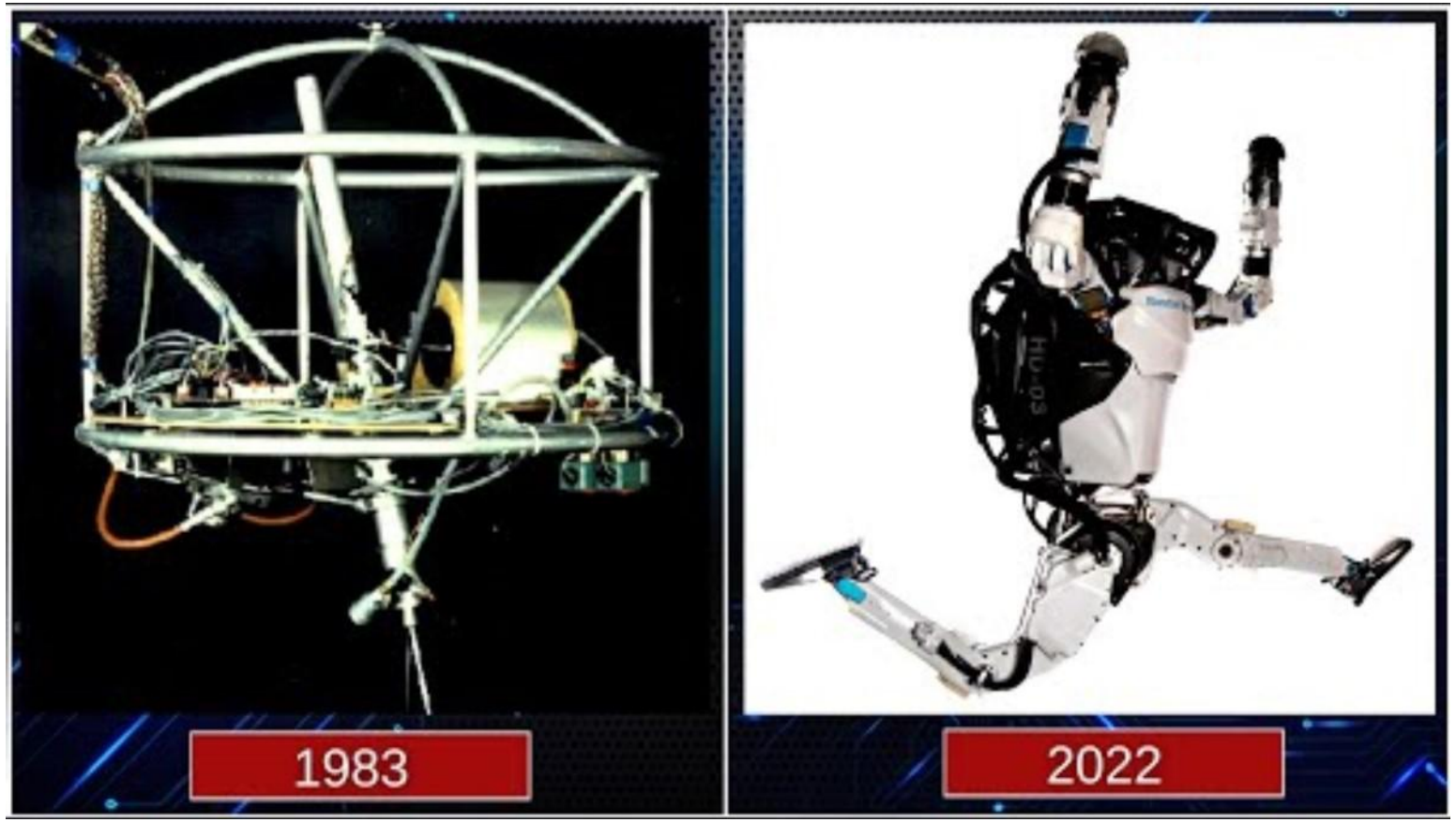


- Course material
  - RVC 3rd Ed. + QUT Robot Academy + lecture notes + files on Canvas

List and discuss some ethical issues within robotics! How should these be handled?

State of the art: Waymo

Robotic Evolution – Boston Dynamics 40 yrs (Hyuandai)



State of the art: DARPA Triage Challenge 2026

https://triagechallenge.darpa.mil

Thank you for your attention!